

MediMind Mid-Defense Report

1. Project Title

MediMind: AI-Assisted Hospital Management System (Django Multi-App Architecture)

2. Team and Submission Context

- Program: BIT 404 Project Work
- Project Type: Full-stack healthcare information system
- Current Stage: Mid-defense (major core + AI roadmap features implemented)

3. Problem Statement

Hospitals and clinics often use disconnected tools for appointments, records, lab reports, prescriptions, and billing. This causes delays, manual errors, and weak traceability. Patients also receive limited proactive guidance from the system.

MediMind addresses this by providing one integrated platform with role-based workflows and practical AI-assisted modules for triage, report interpretation, no-show risk, drug interaction checking, and heart risk screening.

4. Objectives

- Build one integrated role-based hospital workflow platform.
- Reduce operational friction in appointment, lab, payment, and prescription flows.
- Add practical AI modules that assist (not replace) clinical decisions.
- Improve platform security, reliability, and auditability.
- Generate professional legal-compliant documents (receipts, prescriptions, lab reports, appointment PDFs).

5. Scope at Mid-Defense

Completed Core Scope

- Custom authentication with role-based dashboards.
- Doctor profile and schedule management.
- Multi-step appointment booking and queue handling.
- Prescription generation with PDF output.
- Lab test booking, result verification, release, and PDF reports.
- Payment workflow with receipt generation and refund management.
- Room and bed allocation workflows.
- In-app notifications and review module.

Completed AI/Automation Scope

- AI triage (P1-P4 classification).
- AI medical report reader for uploaded reports.
- Drug interaction checker API workflow.
- No-show risk prediction APIs and logs.
- Heart attack risk assessment with ML-first inference and fallback rule engine.
- WhatsApp notifications (Twilio-integrated utility path).

In Progress / Final Phase Scope

- Final analytics polish and export improvements.
- Real-time update layer (WebSocket/channels planned, not yet final).
- Docker/cloud deployment packaging.

6. Methodology

The project follows the NEP method:

- N: Normalize foundation (security, infra, reliability)
- E: Enhance with high-value AI and automation
- P: Production polish and deployment readiness

Development style:

- Modular app-by-app implementation
- Frequent targeted tests after each feature
- Migration-driven schema evolution
- Incremental UI integration through shared base template and role dashboards

7. Current Architecture

- Backend: Django 6.x with server-rendered templates and selected JSON endpoints.
- Database: PostgreSQL (configured as default in settings), with migration-backed models.
- Security: custom middleware for headers and request rate limits, 2FA workflow for sensitive roles.
- Files/Media: PDF/report uploads in media storage.
- Async readiness: Celery/Redis settings prepared.

8. Key Innovation Points for Mid-Defense Demo

- 1. AI Report Reader with structured risk output, abnormal flag extraction, and personalized guidance.
- 2. Heart Risk module using trained ML artifacts from OpenML heart-disease data, with safe fallback logic.
- 3. No-show risk scoring integrated into appointment workflow for operational planning.
- 4. Drug interaction checker endpoint for doctor-side prescription safety checks.

- 5. Professional legal-styled PDF generation across major patient-facing documents.

9. Data Models Snapshot

Representative entities implemented:

- Users and profiles: User, PatientProfile, PatientHealthRecord, PatientVitalLog, TwoFactorCode.
- Clinical: Specialization, Doctor, Shift, DoctorSchedule, DoctorLeave.
- Appointments: Appointment, WaitingList, TriageAssessment, MedicalReportAnalysis, NoShowPredictor.
- Payments: Payment, Refund, PaymentReminder, InsuranceVerification.
- Lab: TestTemplate, TestBooking, TestResult, TestResultItem.
- Prescriptions: Prescription, PrescriptionItem, Refill entities.
- Risk/AI logs: HeartRiskAssessment, NoShowPredictionLog, InteractionCheckLog.

10. APIs and Integration at Mid-Defense

Implemented internal APIs:

- appointments/api/slots/
- appointments/api/no-show-risk/
- heart-risk/api/search-patients/
- drug-checker/api/check-interactions/
- payments/insurance/* JSON endpoints

External integrations configured:

- Twilio WhatsApp API (notifications and OTP support path)
- SMTP email
- Khalti payment key configuration

11. Security and Reliability Work

- Role-gated views and endpoint checks.
- Session/cookie hardening settings.
- Rate limit middleware for authentication-sensitive routes.
- Audit middleware integration.
- 2FA verification and resend workflow.

12. Demo Plan (Mid-Defense)

- 1. Login as admin/receptionist/patient and show role-specific dashboards.
- 2. Book an appointment and show generated payment record.
- 3. Run AI triage with symptom input and explain priority output.
- 4. Upload a sample report in AI report reader and show risk summary + flags + recommendations.

- 5. Perform heart risk assessment and show saved history entries.
- 6. Demonstrate drug interaction API check from doctor flow.
- 7. Download one receipt/report PDF to show legal and branding standard.

13. Results So Far

- End-to-end hospital workflow is functional with integrated data models.
- AI-assisted modules are implemented and visible to users/admin where applicable.
- Admin dashboards now include AI usage visibility for report-reader activity.
- System-level checks and targeted feature tests have been executed during development iterations.

14. Limitations and Risk Disclosure

- AI outputs are decision-support only; not a clinical diagnosis.
- OCR quality and source report formatting can affect report-reader extraction quality.
- Some roadmap items (real-time channels, full cloud deployment hardening) remain for final defense.
- Model performance depends on available training artifacts and data quality.

15. Plan from Mid-Defense to Final Defense

- Finalize analytics/exports and monitoring dashboards.
- Add production deployment stack (Docker + cloud target).
- Complete additional system testing and documentation screenshots.
- Improve observability and benchmark-based performance validation.

16. Conclusion

MediMind has moved beyond a basic HMS into a modular, AI-assisted platform with practical clinical support features, security improvements, and professional documentation outputs. The current build is mid-defense ready and positioned for final production-grade completion in the next phase.